

SOFTWARE REQUIREMENTS SPECIFICATION

Guest & Resident Feedback Collection System

Prepared for: Supun Arcade

Prepared by: Anul Methnula, Business Analyst

Audience: Development Team

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines what the development team must build for the Guest & Resident Feedback Collection System at Supun Arcade. It translates the approved business proposal into concrete functional, data, interface, and technical requirements so the system can be designed, built, and tested without ambiguity.

1.2 Scope

The system allows residents (20 apartments), restaurant customers, pool users, and washroom visitors to submit feedback by scanning a location-specific QR code. Feedback is stored centrally and made available to management through a mobile-friendly web dashboard, with instant WhatsApp alerts for urgent or low-rated submissions.

In scope: guest feedback capture (4 zone types), central data storage, manager dashboard (filter/sort/search/resolve), WhatsApp alerting, and restricted (login-based) dashboard access.

Out of scope for Version 1: payment processing, a native mobile app, and multi-property support (see Section 11).

1.3 Intended Audience

- Development team (frontend, backend, and QA) — primary audience, to build and test the system.
- Project owner / Business Analyst — to verify the build matches business intent.
- Management (the boss) — as the accepting stakeholder for the final system.

1.4 Definitions & Abbreviations

Term	Definition
Zone	One of the four feedback areas: Apartment, Restaurant, Pool, or Washroom.
Location Tag	The identifier captured with each feedback entry, e.g. APT-12, TBL-05, POOL, WC-L1.
SRS	Software Requirements Specification (this document).
Guest	Any resident, restaurant customer, pool user, or washroom visitor submitting feedback.
Manager	The property owner/manager who views and actions feedback via the dashboard.
Urgent Flag	A guest-set flag indicating feedback needs immediate attention, triggering a WhatsApp alert.

1.5 References

- Feedback Collection System — Business Proposal (approved), prepared by the Business Analyst.
- Interactive Prototype — Guest flow and Manager dashboard (HTML prototype, shared with the team).

2. Overall Description

2.1 Product Perspective

This is a new, standalone system. It is not replacing any existing software, since feedback is currently collected informally or not at all. The system consists of a guest-facing web form, a backend API and database, a manager-facing dashboard, and a third-party alerting integration (WhatsApp).

2.2 Product Features Summary

- QR-triggered, location-aware guest feedback form (4 zone types).
- Star rating, category selection, comment, optional contact details, and urgent flag.
- Central database storing all submissions with full location, category, rating, and status.
- Manager dashboard: filter by zone/date range, search, sort, live stats, and status management.
- Instant WhatsApp alert for low ratings (≤ 2 stars) or urgent-flagged feedback.
- Restricted dashboard access via Google account login (single sign-on, session remembered).

2.3 User Classes & Characteristics

User Class	Description	Technical Skill
Guest / Resident	Scans QR, submits feedback via mobile browser	None assumed — must be usable by anyone
Manager / Boss	Views dashboard, filters, resolves issues, receives WhatsApp alerts	Basic smartphone use only
Staff (optional, future)	May be given restricted dashboard views for their zone (e.g. restaurant manager)	Basic smartphone use only
Developer / Admin	Builds, deploys, and maintains the system	Full technical proficiency

2.4 Operating Environment

- Guest form: must run in any modern mobile browser (Chrome, Safari) — no app install required.
- Manager dashboard: mobile-responsive web app, accessible via “Add to Home Screen” bookmark on iOS/Android, and desktop browser.
- Backend: cloud-hosted (see Section 9 for recommended stack).

2.5 Design & Implementation Constraints

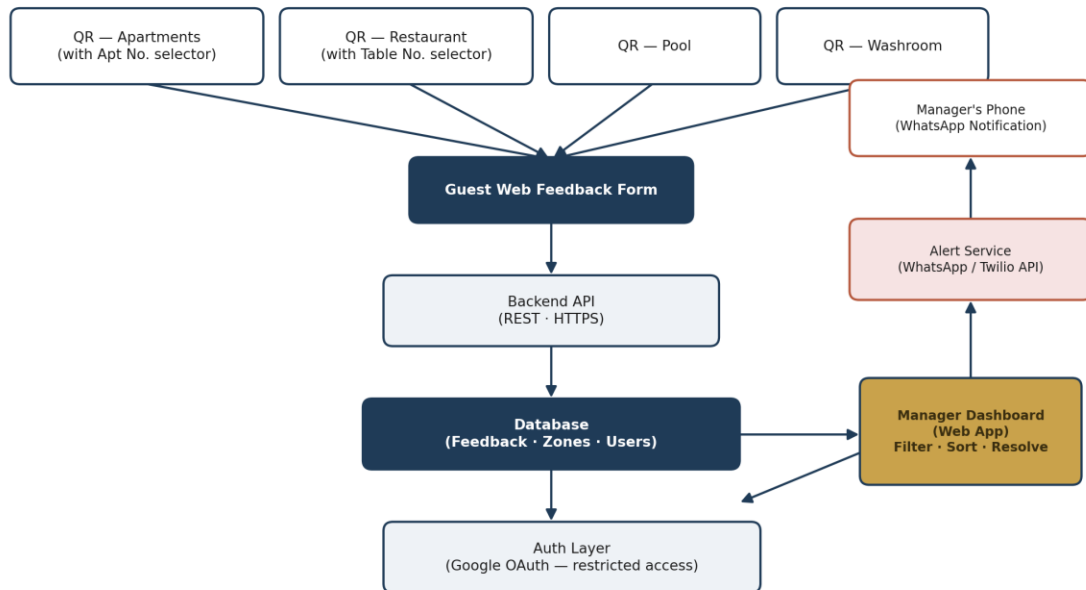
- QR codes are static images; any change of location mapping requires reprinting the affected QR code.
- WhatsApp alerts depend on a third-party API (WhatsApp Business API or Twilio) and its messaging limits/costs.
- Dashboard access requires Google OAuth login; the property must provide approved Google accounts for the manager and any staff.

2.6 Assumptions & Dependencies

- Guests have smartphones with a working camera and mobile internet access at the property.
- The manager checks the dashboard and/or WhatsApp regularly enough to act on alerts.
- QR codes are printed and placed correctly at their intended, matching location.

3. System Architecture

The diagram below shows how data flows from a guest scanning a QR code through to the manager receiving an alert and viewing the dashboard.



- QR codes are zone-specific: Apartments and Restaurant QR codes additionally prompt a quick selector (apartment number / table number) after scanning; Pool and Washroom QR codes require no selector since they are single shared locations.
- The Guest Web Feedback Form calls the Backend API over HTTPS to submit each entry.
- The Backend API writes to the Database and separately triggers the Alert Service when a submission qualifies (rating ≤ 2 or urgent flag = true).
- The Manager Dashboard reads from the Database (via the API) and is protected by the Auth Layer (Google OAuth).

4. Functional Requirements

4.1 QR Code & Location Identification

ID	Requirement
FR-1.1	The system shall generate a unique QR code per zone type: Apartments, Restaurant, Pool, Washroom (minimum 4 codes; more may be added later, e.g. multiple washrooms).
FR-1.2	When the QR for Apartments or Restaurant is scanned, the form shall prompt the guest to select their apartment number (1–20) or table number before proceeding.
FR-1.3	When the QR for Pool or Washroom is scanned, the system shall auto-tag the submission with that fixed location — no guest input required.
FR-1.4	Every submission shall store a resolved Location Tag (e.g. APT-12, TBL-05, POOL, WC-L1) regardless of entry path.

4.2 Guest Feedback Form

ID	Requirement
FR-2.1	The form shall present a 1–5 star rating control as the first input.
FR-2.2	The form shall present category options specific to the scanned zone (see Section 13 Appendix for the full category list per zone), allowing multiple selections.
FR-2.3	The form shall provide an optional free-text comment field.
FR-2.4	The form shall provide optional name and phone number fields for guests willing to be contacted directly.
FR-2.5	The form shall provide an “urgent attention” toggle, which the guest can enable to flag time-sensitive issues.
FR-2.6	The form shall be completable in under 60 seconds and require no login or account creation from the guest.
FR-2.7	On submission, the form shall display a confirmation screen summarizing the rating, category, and location submitted.

4.3 Data Storage

ID	Requirement
FR-3.1	Every submission shall be saved with: timestamp, zone, location tag, category/categories, rating, comment, name (optional), phone (optional), urgent flag, and status.
FR-3.2	Status shall default to “Open” on creation and support transitions to “In Progress” and “Resolved.”
FR-3.3	All submissions shall be retained indefinitely unless a data retention policy is defined later (see Section 7.2).

4.4 Manager Dashboard

ID	Requirement
FR-4.1	The dashboard shall display live summary statistics: total feedback count, average rating, and open request count — recalculated based on active filters.
FR-4.2	The dashboard shall list feedback entries showing: location tag, category, guest name (if provided), comment, exact date/time, star rating, and status.
FR-4.3	The dashboard shall support filtering by zone (Apartments/Restaurant/Pool/Washroom) and by date range (Today/This Week/This Month/All Time).
FR-4.4	The dashboard shall support free-text search across location tag, guest name, and comment.
FR-4.5	The dashboard shall support sorting by Newest First, Oldest First, and Lowest Rating First.
FR-4.6	The manager shall be able to change a feedback entry's status (Open → In Progress → Resolved) directly from the list.
FR-4.7	The dashboard shall be usable on both mobile (as a home-screen web app) and desktop browsers.

4.5 Alerts & Notifications

ID	Requirement
FR-5.1	The system shall send a WhatsApp message to the manager's registered number within 1 minute of any submission with a rating ≤ 2 stars or the urgent flag enabled.
FR-5.2	The alert message shall include the location tag, category, rating, and a shortened summary of the comment.
FR-5.3	WhatsApp alerts shall require no login — they are delivered as a standard WhatsApp message to the manager's existing account.

4.6 Authentication & Access Control

ID	Requirement
FR-6.1	The Manager Dashboard shall be accessible only to pre-approved Google accounts (restricted sharing / OAuth login).
FR-6.2	The guest feedback form shall require no authentication of any kind.
FR-6.3	The system shall remember an authenticated manager's session on their device to avoid repeated logins.

5. Data Requirements

The core data entity is the Feedback record. Supporting entities are Zones and Users (dashboard accounts).

Feedback

Field	Type	Notes
id	UUID / Integer	Primary key, auto-generated
zone	Enum	apartment restaurant pool washroom
location_tag	String	e.g. APT-12, TBL-05, POOL, WC-L1
categories	Array of String	One or more selected categories
rating	Integer (1–5)	Required
comment	Text	Optional
guest_name	String	Optional
guest_phone	String	Optional
urgent	Boolean	Default false
status	Enum	open in_progress resolved
created_at	Timestamp	Auto-set on submission
updated_at	Timestamp	Auto-set on status change

Zones

Field	Type	Notes
id	Integer	Primary key
type	Enum	apartment restaurant pool washroom
label	String	Display name, e.g. “Apartment 12”
requires_selector	Boolean	True for apartment/restaurant; false for pool/washroom

Users (Dashboard Accounts)

Field	Type	Notes
id	Integer	Primary key
name	String	Full name
email	String	Must match an approved Google account
role	Enum	owner manager staff (future use)
phone	String	For WhatsApp alert delivery

6. External Interface Requirements

6.1 User Interfaces

- Guest Feedback Flow: 4 screens — Rating, Category Selection, Comment/Details, Confirmation (see shared prototype for exact layout and interaction).
- Manager Dashboard: Header with property name and alert bell, stat summary row, alert banner, zone/date filters with search and sort, and a scrollable feedback list (see shared prototype).

6.2 Hardware Interfaces

- Guest smartphones with a camera (for scanning QR codes) and a modern mobile browser.
- Printed QR code signage at each of the 4 zone types.

6.3 Software Interfaces

- WhatsApp Business API or Twilio API — for outbound alert messages.
- Google OAuth 2.0 — for restricted manager login.

6.4 Communication Interfaces

- All client-server communication shall use HTTPS.
- The frontend shall communicate with the backend via a REST API returning JSON (see Section 10).

7. Non-Functional Requirements

7.1 Performance

- The guest form shall load within 2 seconds on a standard mobile connection.
- Dashboard filters and search shall update results within 1 second for up to 10,000 feedback records.

7.2 Security & Privacy

- Guest-submitted data (name, phone, comments) shall not be exposed publicly or to unauthorized accounts.
- Dashboard access shall be restricted to explicitly approved Google accounts (see FR-6.1).
- A data retention/deletion policy should be defined before launch (e.g. retain feedback data for 24 months).

7.3 Usability

- The guest form shall require no instructions, login, or app installation to use.
- The dashboard shall be operable on a phone screen without horizontal scrolling.

7.4 Reliability & Availability

- The system shall target 99% uptime for the guest form and dashboard.
- Failed WhatsApp alert deliveries shall be retried at least once.

7.5 Scalability

- The system shall support adding new apartments, tables, or additional washrooms without code changes (configuration-driven zone list).

7.6 Compatibility

- The guest form and dashboard shall function correctly on the latest two major versions of Chrome and Safari (mobile and desktop).

7.7 Maintainability

- Category lists per zone shall be configurable without a full redeployment, to allow future adjustments.

8. Use Case Scenarios

UC-1: Guest submits feedback at the restaurant

A customer finishes their meal, scans the Restaurant QR code, selects Table 05, rates their experience 4 stars, selects “Food Taste” and “Speed,” adds a short comment, and submits. The system stores the entry and it appears instantly on the manager's dashboard under the Restaurant filter.

UC-2: Guest reports an urgent maintenance issue

A resident in Apartment 9 scans the Apartments QR code, selects Apartment 9, rates the experience 1 star, selects “Maintenance,” writes a comment about a broken air conditioner, and enables the urgent toggle. The system immediately sends a WhatsApp alert to the manager, in addition to storing the entry as “Open.”

UC-3: Manager reviews and resolves feedback

The manager receives the WhatsApp alert, taps through to the dashboard (already logged in from a prior session), filters by Apartments, finds the Apartment 9 entry, dispatches maintenance staff, and once fixed, marks the entry “Resolved” from the dashboard.

9. Recommended Technology Stack

This is a recommended starting stack; the development team may substitute equivalent tools as long as the functional and non-functional requirements above are met.

Layer	Recommended Option
Guest Form & Dashboard (Frontend)	React or plain HTML/CSS/JS — mobile-first, responsive design
Backend API	Node.js with Express, or a managed backend such as Firebase Functions
Database	PostgreSQL (self-managed) or Firebase Firestore (managed, faster to launch)
Authentication	Google OAuth 2.0 (via Firebase Auth or a standard OAuth library)
Alerts	WhatsApp Business API, or Twilio WhatsApp API as a simpler alternative
Hosting	Vercel or Netlify (frontend); Firebase / Render / Railway (backend)
QR Code Generation	Any standard QR generator library, pointing to unique form URLs per zone

10. API Endpoint Specification

Method	Endpoint	Purpose	Auth Required
POST	/api/feedback	Submit a new feedback entry	No
GET	/api/feedback	List feedback entries (supports zone, date range, search, sort query params)	Yes
PATCH	/api/feedback/:id/status	Update the status of a feedback entry	Yes
GET	/api/stats	Return summary statistics (count, average rating, open count) for given filters	Yes
GET	/api/zones	List configured zones and whether they require a selector	Yes

11. Out of Scope / Future Enhancements

- Native mobile app (Version 1 uses a mobile-responsive web app / home-screen bookmark only).
- Payment or billing integration.
- Multi-property support (system is scoped to this single property for Version 1).
- Role-based views for individual staff (e.g. restaurant manager seeing only restaurant feedback) — noted as a future enhancement in the Users data model.
- Sinhala/English language toggle on the guest form — recommended as a near-term enhancement given the local guest base.

12. Acceptance Criteria

- A guest can complete and submit feedback from any of the 4 zone QR codes in under 60 seconds without creating an account.
- Every submission appears on the manager dashboard with the correct location tag within a few seconds.
- A 1–2 star rating or urgent-flagged submission triggers a WhatsApp alert to the manager within 1 minute.
- The manager can filter by zone and date range, search, sort, and mark an entry resolved, all from a mobile device.
- The dashboard cannot be opened by a Google account that has not been explicitly approved.

13. Appendix — Feedback Categories per Zone

Zone	Categories
Apartments	Staff Service, Maintenance, Cleanliness, Security, Noise, Other
Restaurant	Food Taste, Staff Service, Cleanliness, Speed, Price, Other
Pool	Cleanliness, Safety, Staff Service, Equipment, Other
Washrooms	Cleanliness, Supplies, Maintenance, Other